Daily Review



1	The diamond is the … symbol.	Decision
2	x3? makes what kind of loop?	A fixed loop
3	Every delay box needs its…	Unit
4	A loop going up needs an…	Arrow
5	An input becomes which box?	A decision

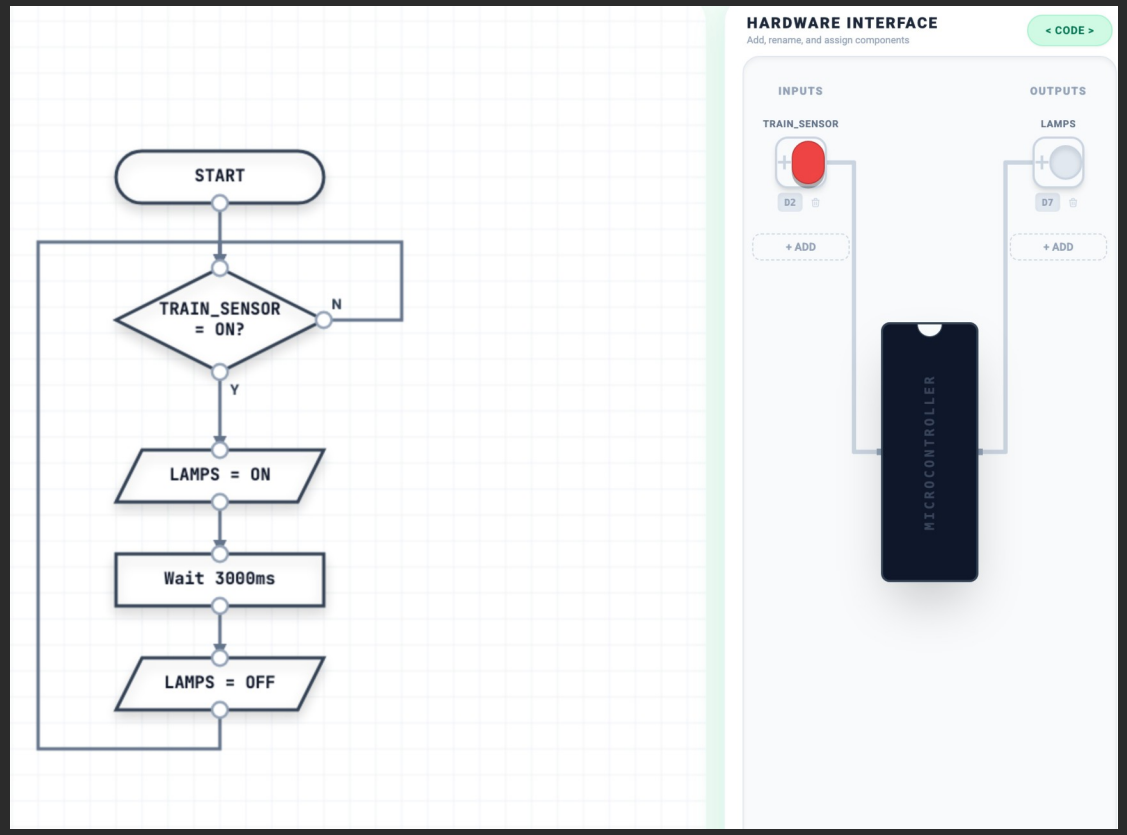
Reading Programs

The exam writes a flowchart as lines. Can you read them?



National 5 Engineering Science

Here is the flowchart. Written as lines, which line is the delay?



A flowchart in NoSyntaxSim

PART 1

Reading the Exam's Program

Line by line, box by box.

By the end of this part…



1. I can say what each line of an exam program does
2. I can find a line from what it does
3. I can say how many times a counter loop runs

The Words the Exam Uses



switch on 7

pin 7 on

pause 500

wait 500 ms = 0.5 s

main: label_1:

a name for a line

goto main

jump to that line

if … then label_1

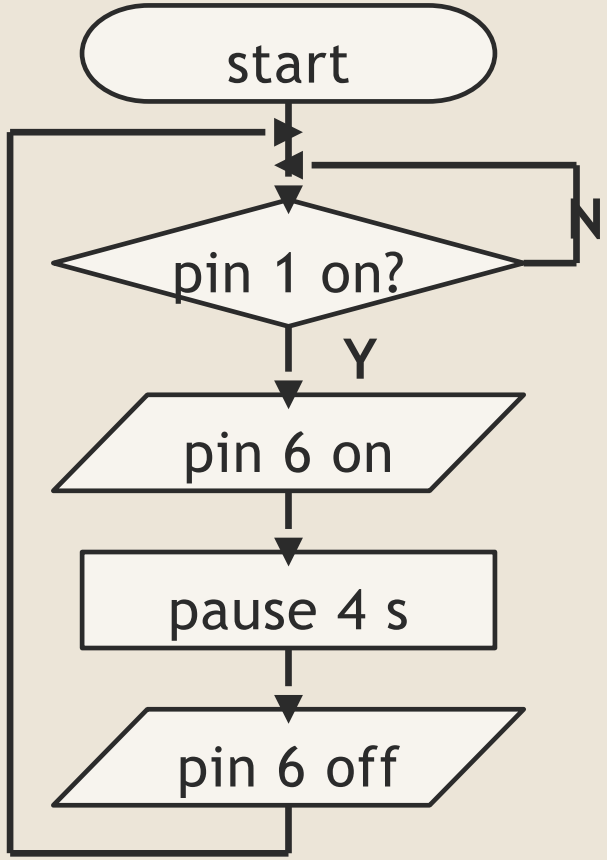
a decision

let count = …

a counter

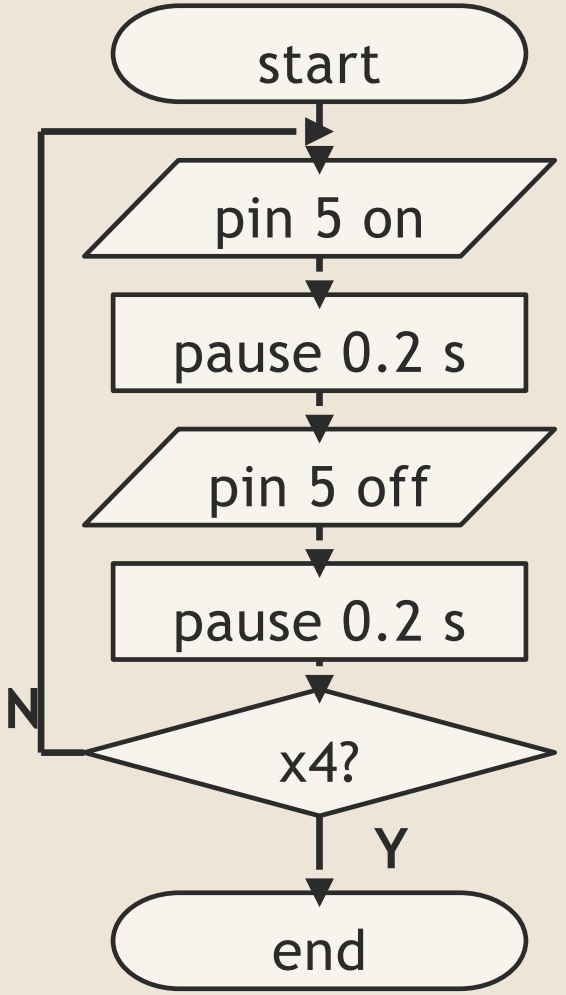
One Program, One Flowchart

```
1    main:      if input1 is off then main
2               switch on 6
3               pause 4000
4               switch off 6
5               goto main
```



Counting: The Fixed Loop in Code

```
1   main:       let count = 0
2   label_1:    switch on 5
3               pause 200
4               switch off 5
5               pause 200
6               let count = count + 1
7               if count = 4 then label_2
8               goto label_1
9   label_2:    end
```



Read It




```
1    main:      if input2 is off then main
2               let count = 0
3    label_1:   switch on 7
4               pause 1500
5               switch off 7
6               let count = count + 1
7               if count = 8 then label_2
8               goto label_1
9    label_2:   pause 2500
10              goto main
```

1 A line with a delay?

4 or 9

2 Times pin 7 turns on?

8 times

3 Lines 3-8: what loop?

Fixed

4 What does line 5 do?

Switches pin 7 off

5 How long is line 9?

2.5 s

Model Answer: The Function of a Line (1 mark)



```
1    main:    if input1 is off then main
2             switch on 6
3             pause 4000
4             switch off 6
5             goto main
```

Describe the function of line 5.

Answer (1)

It sends the program back to line 1, so it repeats forever.

Scores 0

“It goes to main.”

Checkpoint



1	pause 750 waits how long?	0.75 s
2	What does 'goto main' make?	A continuous loop
3	Why is a label needed?	To name a line — **so goto can jump there**

Plenary: Part 1



1	switch off 4 means…	pin 4 off
2	pause is measured in…	Milliseconds
3	if count = 5 then label_2 is a…	Decision
4	let count = count + 1 does what?	Adds 1 to the count
5	'Goes to main' on its own scores…	0 marks

Look back — by the end of this part…



1. I can say what each line of an exam program does
2. I can find a line from what it does
3. I can say how many times a counter loop runs

Daily Review



1	pause 1500 waits…	1.5 s
2	goto label_1 jumps to…	The line named label_1
3	let count = 0 does what?	Sets the count to 0
4	A line with a delay starts with…	pause
5	'Goes to main' alone scores…	0

PART 2

When It Goes Wrong

It should beep 10 times. It never stops. Why?

By the end of this part…



1. I can trace a counter loop, pass by pass
2. I can explain why a program does not repeat
3. I can turn a flowchart into a program

Trace It, Pass by Pass




```
1    main:       let count = 0
2    label_1:    switch on 4
3                let count = count + 1
4                switch off 4
5                if count = 3 then label_2
6                goto label_1
7    label_2:    end
```

Run	count	= 3?	line
1	1	no	2
2	2	no	2
3	3	yes	7

Model Answer: Why It Does Not Repeat (2 marks)



```
1   main:       let count = 0
2   label_1:    switch on 7
3               pause 300
4               switch off 7
5               pause 300
6               let count = count + 1
7               if count = 10 then label_2
8               goto main
9   label_2:    switch on 5
10              end
```

Explain why lines 2-7 don't run 10 times.

Cause (1)

Line 8 goes back to line 1,

Effect (1)

so the count resets: never 10.

Watch Out: Loop Back Below the Count

THE MISTAKE

- 8 goto main
- The count goes back to 0.
- It never reaches 10.

THE CORRECT ANSWER

- 8 goto label_1
- The count keeps going up.
- It stops at 10.

Spot the Mistake



```
1   main:       let count = 0
2   label_1:    switch on 6
3               pause 500
4               switch off 6
5               pause 500
6               if count = 3 then label_2
7               goto label_1
8   label_2:    switch on 7
9               end
```

It should

flash pin 6 three times, then pin 7 on

It does

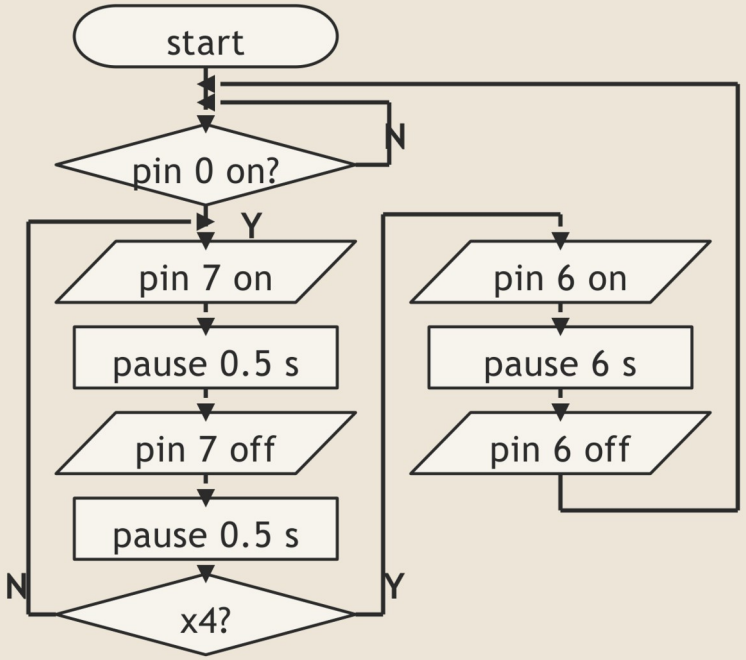
flash pin 6 for ever

Your job

find the missing line

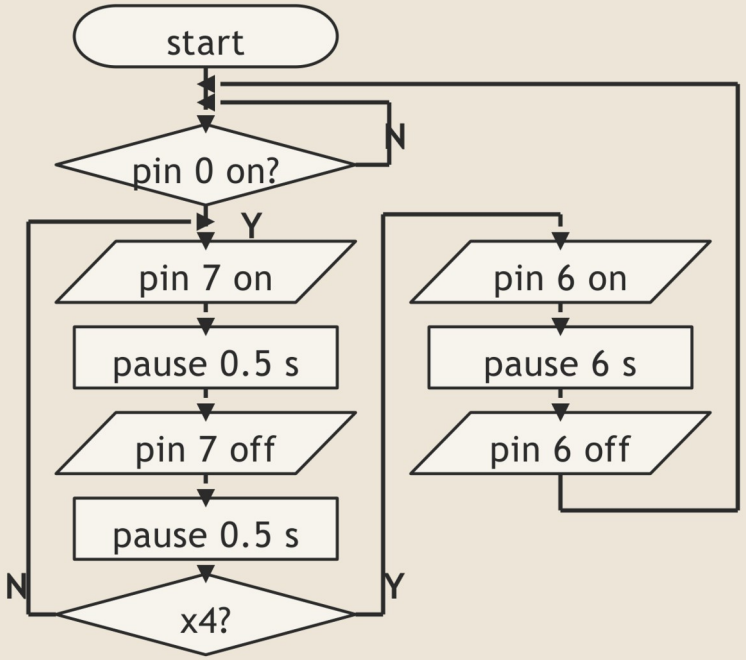
Flowchart to Program (1 of 2)

```
1   main:       if input0 is off then main
2               let count = 0
3   label_1:    switch on 7
4               pause 500
5               switch off 7
6               pause 500
7               let count = count + 1
8               if count = 4 then label_2
9               goto label_1
```



Flowchart to Program (2 of 2)

```
10  label_2:    switch on 6
11               pause 6000
12               switch off 6
13               goto main
```



Your Turn: Program to Flowchart



```
1   main:    if input3 is off then main
2            switch on 5
3            switch on 4
4            pause 2000
5            switch off 5
6            pause 1000
7            switch off 4
8            goto main
```

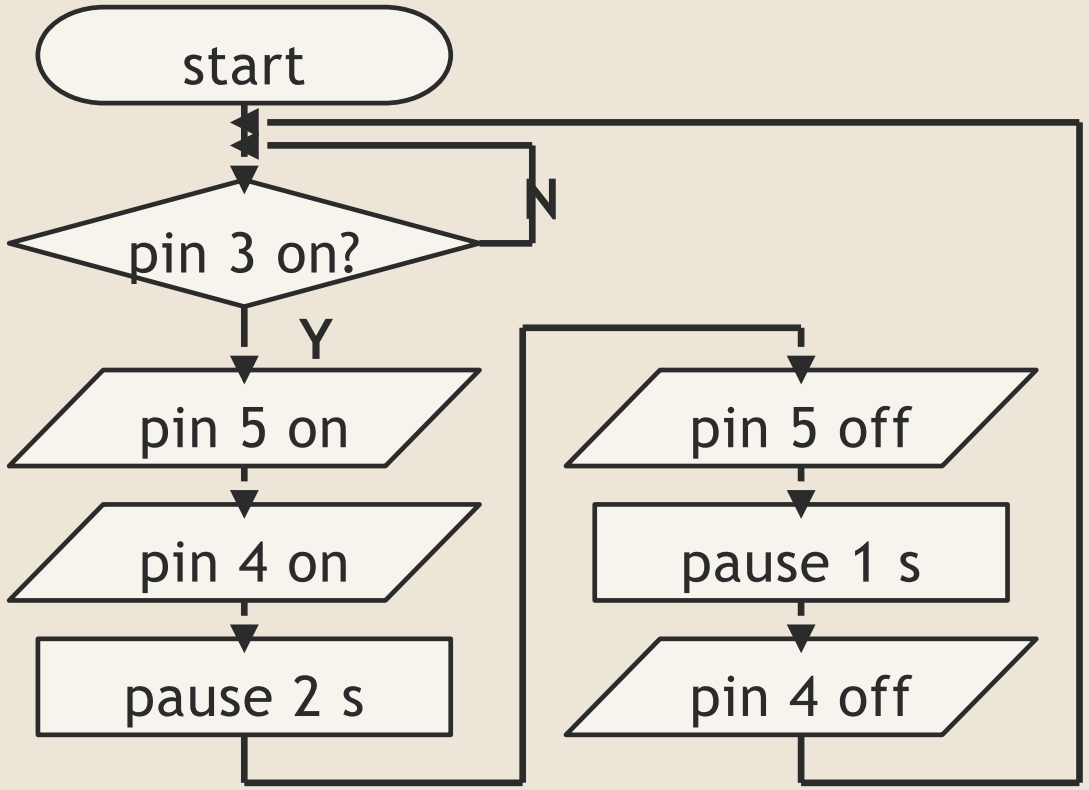
Draw

the flowchart for this program

Check

every pin, every unit, every arrow

Your Turn: The Answer



Your Turn: Flowchart to Program

Write

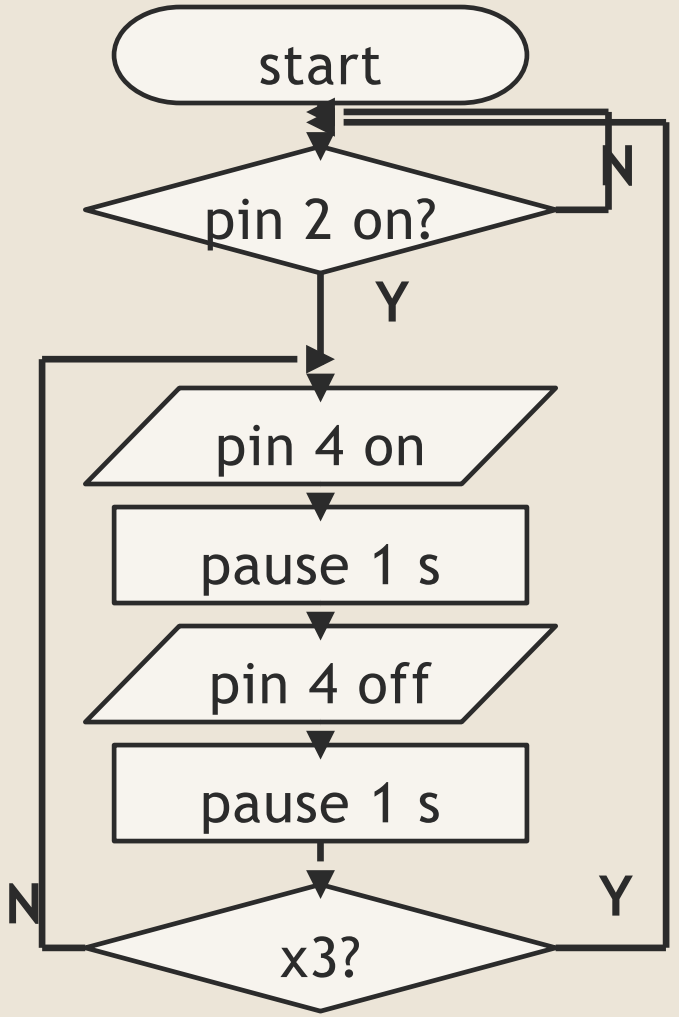
this flowchart as a program

One line

for each box

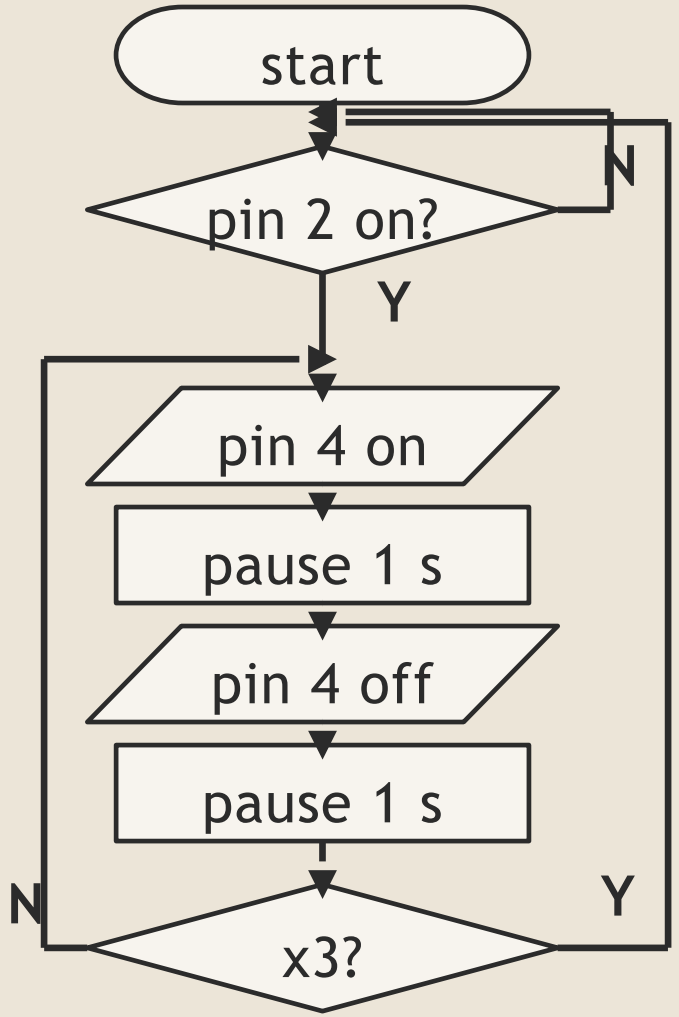
Check

labels, ms, the goto



Your Turn: The Program

```
1   main:       if input2 is off then main
2               let count = 0
3   label_1:    switch on 4
4               pause 1000
5               switch off 4
6               pause 1000
7               let count = count + 1
8               if count = 3 then main
9               goto label_1
```



Past Papers: Reading Programs

A microcontroller based system is used to detect the bottles.

The program used to count six bottles is shown below.

line	program	
1	main:	let count = 0
2	check:	if input2 is off then check
3		let count = count + 1
4		if count = 6 then label_1
5		goto check
6	label_1:	switch on 7
7		pause 500
8		switch off 7
9		goto main

(c) State the **line number** that contains a time delay.

_____

2023 Q9(c)

2. A microcontroller program is shown.

line	program	
1	main:	if input1 is on then label_1
2		goto main
3	label_1:	let count = 0
4	label_2:	switch on pin 7
5		pause 500
6		switch off pin 7
7		pause 500
8		let count = count + 1
9		if count = 15 then label_3
10		goto label_2
11	label_3:	end

(a) State a line number that contains a time delay.

Line number _____

(b) State the number of times that pin 7 will turn on and then off.

_____

(c) State the name of the type of loop used to control the number of times pin 7 turns on and off.

2025 Q2

A program used in a different control system is listed below.

line	*program*	
1	main:	let count = 0
2	label_1:	switch on 4
3		switch on 5
4		pause 600
5		switch off 4
6		switch off 5
7		pause 600
8		let count = count + 1
9		if count = 20 then label_2
10		goto main
11	label_2:	if Input0 is on then label_3
12		goto label_2
13	label_3:	switch on 7
14		pause 3000
15		switch off 7
16		goto main

(b) Describe the function of **line 16** in the program.

_____

_____

Lines 2 to **9** should repeat twenty times before moving on to **line 11**.

During testing an electronic engineer found that this did not happen.

(c) Explain why **lines 2** to **9** did not repeat twenty times before moving on to **line 11**.

2022 Q9(b)(c)

Past Papers: Answers



1	2023 Q9(c)	Line 7
2	2025 Q2(a), (b)	Line 5 or 7; 15 times
3	2025 Q2(c)	Fixed (finite) loop
4	2022 Q9(b)	Back to line 1 — **so it repeats forever**
5	2022 Q9(c) (2)	Loops to line 1 (1) — **so count resets (1)**

Past Papers: Find Them All

Past papers

Year	Question
2022	Q9(b), (c)
2023	Q9(c)
2025	Q2

By topic

Checkpoint



1. A count never reaches its limit. Look for?

   A goto above let count = 0

2. No count + 1 line: what happens?

   It loops for ever

3. Why must the loop go to label_1?

   Below the reset — **so the count keeps going**

Plenary: Part 2



1	Explain questions need a cause and…	An effect
2	The words that join them?	So …
3	x4? becomes how many lines?	Three
4	The arrow back up becomes…	A goto line
5	The N loop of line 1 goes to…	main (itself)

Look back — by the end of this part…



1. I can trace a counter loop, pass by pass
2. I can explain why a program does not repeat
3. I can turn a flowchart into a program

Daily Review



1	pause 2000 waits…	2 s
2	A goto above the count reset makes…	A loop that never ends
3	Explain questions need…	Cause and effect
4	switch on 3 means…	pin 3 on
5	The flowchart's x3? becomes…	Counter lines

PART 3

Two More Ways to Write It

The booklet's pseudocode, then real Arduino C.

By the end of this part…



1. I can describe what setup() and loop() do
2. I can match a line of Arduino C to what it does
3. I can write pseudocode for a simple program

Three Ways to Write One Program

The exam

```
main:
switch on 5
pause 500
switch off 5
pause 1000
goto main
```

The booklet

```
loop
high buzzer
pause 0.5 seconds
low buzzer
pause 1 second
end loop
```

Arduino C

```
void loop() {
digitalWrite(5, HIGH);
delay(500);
digitalWrite(5, LOW);
delay(1000);
}
```

The Standard Layout



setup()

runs once: set each pin as input or output

loop()

runs for ever: the control program

```
void setup() {
  pinMode(5, OUTPUT);
}

void loop() {
  // the program
}
```

The Commands You Need



pinMode(4, OUTPUT);

pin 4 is an output (in setup)

digitalWrite(4, HIGH);

pin 4 on · LOW: off

delay(250);

wait 250 ms

digitalRead(2)

read the input on pin 2

Let the Simulator Write It

Flowchart in

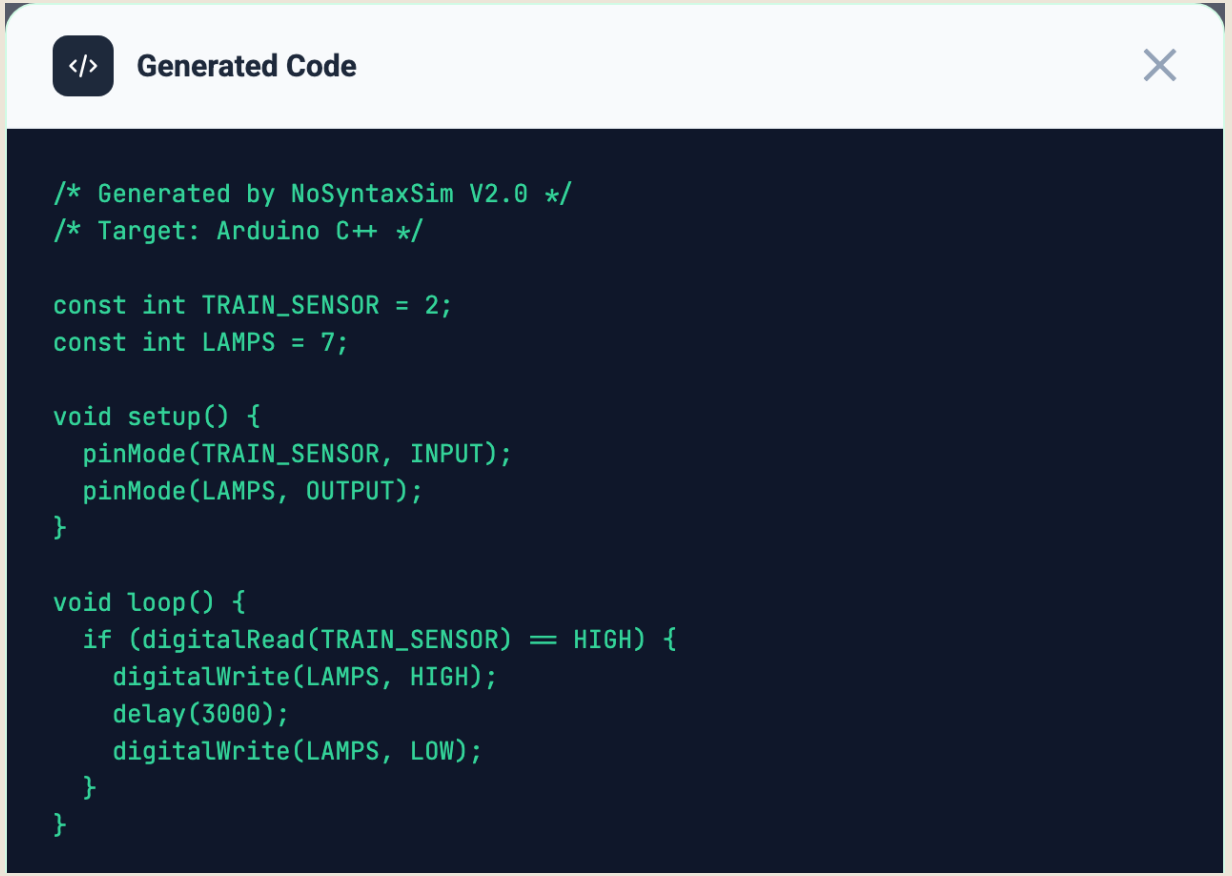
Arduino C out

The decision

if (digitalRead(...) == HIGH)

The wait

delay(3000)



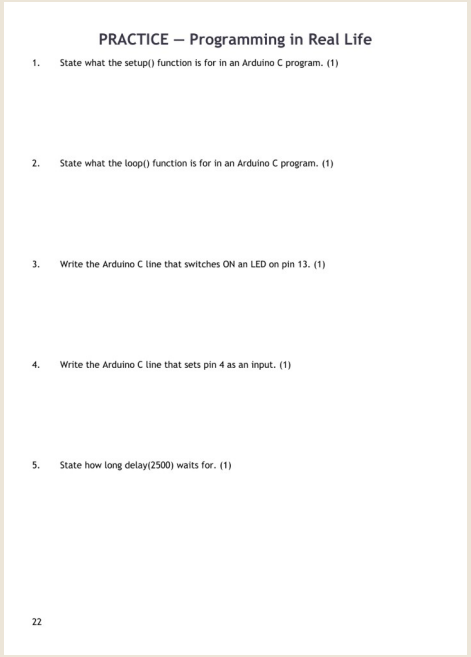
What Does It Do?

1	pinMode(3, INPUT);	Sets pin 3 as an input
2	digitalWrite(7, HIGH);	Switches pin 7 on
3	delay(250);	Waits 0.25 s
4	digitalWrite(7, LOW);	Switches pin 7 off
5	digitalRead(2)	Reads the input on pin 2

PRACTICE: Programming in Real Life

Booklet pages 22-23

Q1 to Q6, on your own.
Q7 and Q8 are for Lesson 4.



PRACTICE — Programming in Real Life

1. State what the setup() function is for in an Arduino C program. (1)

2. State what the loop() function is for in an Arduino C program. (1)

3. Write the Arduino C line that switches ON an LED on pin 13. (1)

4. Write the Arduino C line that sets pin 4 as an input. (1)

5. State how long delay(2500) waits for. (1)

22

12 min

Programming in Real Life: Answers



1	Q1 setup()	Runs once; sets the pins
2	Q2 loop()	Runs for ever: the main program
3	Q3 and Q4	digitalWrite(13, HIGH); pinMode(4, INPUT);
4	Q5 delay(2500)	2.5 s
5	Q6 (4 marks)	for count = 1 to 4 … next count, then end

Plenary: Part 3

1	setup() runs how often?	Once
2	loop() runs…	For ever
3	delay(1000) waits…	1 s
4	The booklet's pause 1 second in the exam?	pause 1000
5	Which tool writes Arduino C for you?	NoSyntaxSim Generate Code

Extension

EXTRA PRACTICE



1. Line 8 of the 4-beep program says goto main. Effect?

   Beeps for ever

2. for (int i = 0; i < 5; i++) runs how often?

   5 times

3. Why does Arduino C need no 'goto main'?

   loop() repeats — **so it is built in**

Look back — by the end of this part…



1. I can describe what setup() and loop() do
2. I can match a line of Arduino C to what it does
3. I can write pseudocode for a simple program

LOOKING AHEAD

Lesson 4: Build, Test and Fix

Put it on the Uno. Then find what is wrong with it.

Course Specification

Digital electronic control systems

- using suitable commands (high, low, for…next, if…then, pause, end (or their equivalents)) to design programs to solve simple control problems, involving time delays and continuous and fixed loops

Credits (1 of 2)

Title slide (NoSyntaxSim): NoSyntaxSim V2 screenshot (Generate Code). Simulator by Peter Strain, NoStrainSims (sites.google.com/view/nostrainsim), free and open to all.

The big question (NoSyntaxSim): NoSyntaxSim V2 screenshot. Simulator by Peter Strain, NoStrainSims (sites.google.com/view/nostrainsim), free and open to all.

Model Answer: The Function of a Line (2022 Q9(b) MI): © Qualifications Scotland (SQA). N5 Engineering Science marking instructions: 2022 Q9(b)(c), 2023 Q9(c), 2025 Q2.

Model Answer: Why It Does Not Repeat (2022 Q9(c) MI): © Qualifications Scotland (SQA). N5 Engineering Science marking instructions: 2022 Q9(b)(c), 2023 Q9(c), 2025 Q2.

Watch Out (2022 Q9(c) MI): © Qualifications Scotland (SQA). N5 Engineering Science marking instructions: 2022 Q9(b)(c), 2023 Q9(c), 2025 Q2.

Credits (2 of 2)

Past Papers: Reading Programs: © Qualifications Scotland (SQA). N5 Engineering Science 2022 Q9(b)(c), 2023 Q9(c) and 2025 Q2, question papers and marking instructions.

Let the Simulator Write It: NoSyntaxSim V2 screenshot (Generate Code). Simulator by Peter Strain, NoStrainSims (sites.google.com/view/nostrainsim), free and open to all.

PRACTICE: Programming in Real Life (booklet page): Booklet 4b pages by R Stewart.

R Stewart - October 2026

Course Specification: © Qualifications Scotland (SQA). Course spec v2.0, p.5